

The Proios Learning Route

Learn

Understand

Recall

Revise

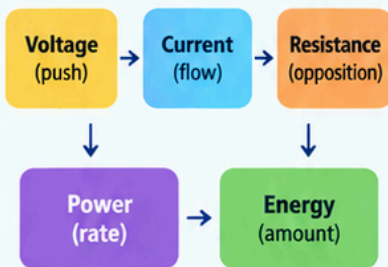
Apply

Test

Improve

1 THE BIG PICTURE

Electricity is a group of effects involving electric charge and electrical energy.



Think of it as a chain — each part affects the next.

2 10 KEY CONCEPTS

Electricity	Effects involving charge and energy
Electron	Tiny negatively charged particle
Circuit	Complete path for operation
Voltage (V)	Electrical push / potential difference
Current (A)	Rate of charge flow
Resistance (Ω)	Opposition to current
Power (W)	Rate of energy transfer
Energy (kWh)	Amount used/transferred
AC	Changes direction
DC	One direction (basic model)

3 THE 4 FORMULAS

Ohm's Law

$$V = I \times R$$

$$I = V \div R \quad R = V \div I$$

Power

$$P = V \times I$$

Example: $12 \text{ V} \times 2 \text{ A} = 24 \text{ W}$

Energy

$$\text{Energy} = \text{Power} \times \text{Time}$$

Example: $2 \text{ kW} \times 3 \text{ h} = 6 \text{ kWh}$

Memory shortcut:

V-I-R = electricity relationship
 $P = V \times I$ = power
 kWh = energy

4 UK ELECTRICITY BASICS

Mains supply

230 V AC
50 Hz



Conductor colours

	Live
	Neutral
	Earth

Live / Neutral / Earth

	Live	Dangerous supply conductor.
	Neutral	Return path for current.
	Earth	Protective conductor/connection.

Earth ≠ Neutral
Keep repeating this until you understand it.

5 PROTECTION – DIFFERENT JOBS

	Fuse	Interrupts excessive current when it operates.
	Circuit breaker	Automatically interrupts certain abnormal conditions.
	RCD	Detects residual current and disconnects under specified conditions.
	Earthing	Provides a protective fault-current path.
	Insulation	Prevents unwanted contact/current paths.

No single device protects against everything.
Safety uses layers.

6 SAFETY – STOP METHOD

S	See The warning sign
T	Think Before touching
O	Observe From a safe position
P	Professional help If necessary

Common hazard signs

- Burning smell
- Melted plastic
- Exposed cables
- Unusual buzzing
- Repeated tripping
- Scorch/burn marks
- Damaged cables
- Sparks
- Hot plugs/sockets

7 DON'T CONFUSE! (5 KEY PAIRS)

Voltage	≠	Current
Power	≠	Energy
Earth	≠	Neutral
Fuse	≠	RCD
Switch off	≠	Proven safe

8 WHAT NOT TO MEMORISE

Tier 1 – Must know

- Voltage, Current, Resistance
- Ohm's Law, Power, Energy
- Circuit, AC/DC, Live/Neutral/Earth
- Fuse/RCD, Basic hazards

Tier 2 – Understand

- Consumer unit, Conductors
- Insulation, Mains supply journey
- UK plug terminology

Tier 3 – Reference

- Detailed glossary, Extended explanations
- O&A, Full test material

9 30-SECOND REVISION CARD

V	Voltage = push
A	Current = flow
Ω	Resistance = opposition
W	Power = rate
kWh	Energy = amount

UK domestic supply
230 V AC | 50 Hz

Live Neutral Earth

Fuse = excess current RCD = residual current

Short Version ⚡ UK ADULT ELECTRICITY - LEVEL 1 by eB Publications, 19 September 2026
- Certified by Unios - Unios Award or Non-Formal Certification (NFCert) for portfolios, CVs, etc.
| Information Overload Solutions Version (iOS Version)

UK ADULT ELECTRICITY
LEVEL 1 — iOS VERSION
Information Overload Solutions Version (Short Version)

Purpose:
Understand the basics. Reduce information overload. Recognise hazards. Know when professional help is needed.

LEARNING ROUTE

Learn → Understand → Recall → Revise → Apply → Test → Improve

1. THE BIG PICTURE

Electricity → Circuit → Voltage → Current → Resistance → Power → Energy → Safety

Simple mental model:

Voltage = electrical push / potential difference

↓

Current = rate of charge flow

↓

Resistance = opposition to current

↓

Power = rate of energy transfer

↓

Energy = amount transferred or used

2. 10 KEY CONCEPTS

3. Electricity — electrical phenomena involving charge and energy

4. Electron — tiny negatively charged particle

5. Circuit — suitable path for electrical operation

6. Voltage — electrical potential difference

- 7. Current — rate of electric charge flow
- 8. Resistance — opposition to electric current
- 9. Power — rate of energy transfer
- 10. Energy — amount of energy transferred or used
- 11. AC — Alternating Current; periodically changes direction
- 12. DC — Direct Current; basic model of current in one direction

13. THE KEY FORMULAS

Ohm's Law:

$$V = I \times R$$

$$I = V \div R$$

$$R = V \div I$$

Example:

$$12 \text{ V} \div 6 \text{ } \Omega = 2 \text{ A}$$

Power:

$$P = V \times I$$

Example:

$$12 \text{ V} \times 2 \text{ A} = 24 \text{ W}$$

Energy:

$$\text{Energy} = \text{Power} \times \text{Time}$$

Example:

$$2 \text{ kW} \times 3 \text{ hours} = 6 \text{ kWh}$$

4. UK ELECTRICITY BASICS

UK domestic supply:

230 V nominal AC

50 Hz

Modern UK conductor colours:

Brown = Live

Blue = Neutral

Green/yellow = Earth

Remember:

Earth $\neq$ Neutral

Live can present dangerous voltage.

Neutral must not automatically be assumed safe to touch.

Earth is primarily protective.

5. PROTECTION — DIFFERENT JOBS

Fuse:

Interrupts excessive current when the fuse operates.

Circuit breaker:

Automatically interrupts certain excessive-current or fault conditions.

RCD — Residual Current Device:

Detects residual current and disconnects under specified conditions.

RCBO:

Combines functions associated with overcurrent and residual-current protection.

Earthing:

Provides a protective fault-current path.

Insulation:

Helps prevent unwanted electrical contact and current paths.

Key principle:

No single device protects against everything.

Electrical safety uses layers.

6. SAFETY — STOP METHOD

S — See

See the warning sign.

T — Think

Think before touching.

O — Observe

Observe from a safe position.

P — Professional help

Get appropriate professional help when necessary.

COMMON HAZARDS

- Burning smell
- Scorch or burn marks
- Melted plastic
- Damaged cables
- Exposed conductors
- Sparks
- Unusual buzzing
- Hot plugs or sockets
- Repeated protective-device tripping

7. DON'T CONFUSE THESE

Voltage $\neq$ Current

Power $\neq$ Energy

Earth $\neq$ Neutral

Fuse $\neq$ RCD

Switch off $\neq$ Proven safe

8. WHAT NOT TO MEMORISE

TIER 1 — MUST KNOW

- Voltage
- Current
- Resistance
- Ohm's Law

- Power
- Energy
- Circuit
- AC and DC
- Live, Neutral and Earth
- Fuse and RCD
- Basic electrical hazards

TIER 2 — UNDERSTAND

- Consumer unit
- Conductors
- Insulation
- UK mains supply journey
- UK plug terminology
- Purpose of different protective devices

TIER 3 — REFERENCE

- Detailed glossary
- Extended explanations
- Additional questions and answers
- Full supporting material

9. 30-SECOND REVISION CARD

Voltage = push / potential difference

Current = rate of charge flow

Resistance = opposition to current

Power = rate of energy transfer

Energy = amount transferred or used

Circuit = suitable electrical path

UK domestic supply = approximately 230 V AC, 50 Hz

Brown = Live

Blue = Neutral

Green/yellow = Earth

Fuse = excessive-current protection

RCD = residual-current protection

Safety = do not improvise

10. IOS LEARNING ROUTE

11. UNDERSTAND

Read the 30-second revision card.

12. RECALL

Cover the answers and explain:

V, A, Ω , W and kWh.

13. APPLY

Practise simple calculations:

$$12 \div 6 = 2 \text{ A}$$

$$12 \times 2 = 24 \text{ W}$$

$$2 \times 3 = 6 \text{ kWh}$$

14. SAFETY

Practise recognising hazards using STOP.

15. TEST

Attempt the Level 1 assessment without looking at the material.

16. IMPROVE

Only revisit the areas you missed.

FINAL IOS MEMORY CARD

Electricity = charge + electrical energy

Voltage = potential difference

Current = charge flow rate

Resistance = opposition

Power = energy transfer rate

Energy = amount transferred or used

Circuit = suitable path

UK domestic electricity = approximately 230 V AC, 50 Hz

Brown = Live

Blue = Neutral

Green/yellow = Earth

Protection = fuses + circuit breakers + RCDs + earthing + insulation

SAFETY:

Level 1 knowledge is NOT electrical competence.

Do not treat this knowledge as permission to work on live mains electricity.

When work is beyond your competence, use an appropriately competent or qualified professional.

iOS PRINCIPLE:

Learn less at one time.

Understand more.

Recall what matters.

Apply safely.

Avoid information overload.